

**STTHK3133 HUMAN FACTOR ENGINEERING**  
**Semester 1 Session 2025/2026 (A251)**

# Assignment 3

**Mental Workload and Automation**  
**Human Performance and Awareness**  
**Ergonomics and Implications**

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|-----------|----------------|
| Type:     | Group          |
| Mark:     | 20%            |
| Deadline: | <b>Week 13</b> |

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1. Based on the dashboard that you have designed in Assignment 2, evaluate the effectiveness of your dashboard design by conducting a structured assessment focusing on:
  - a. Mental workload & automation
  - b. Human performance & awareness
  - c. Ergonomics & design implications
2. Recruit at least five (5) participants (representing prospective users of the dashboard) to interact with your dashboard prototype.
3. Conduct a usability and workload evaluation with real users
4. Analyze how the dashboard supports efficiency and awareness
5. Evaluate ergonomic suitability, considering layout, visibility, colour, font, control placement, cognitive load, etc. Report the ergonomic implications of your design decisions.
6. Analyze findings by connecting results to the three subtopics (how design affects workload, awareness, and ergonomic usability).
7. Propose design improvements based on the evaluation conducted, supported by human factors theory.

## ASSESSMENT

Assessment will be based on problem solving skills. Students are required to submit documents on UUM Online Learning by **Week 13**.